lit2026 W12: flagged ECDLP claims

Zero-minor guess-and-determine (arXiv 2607.09814) against rho.negation; Hu ePrint 2025/015.

2026-10-11 · audit workstream W12 · run `run-20261012T0009Z` · solver commit `c70d57ec7` · L0 host · counted field operations.

Verdict

Zero-minor method (Mahalanobis; 1703.07544 ⟶ 2310.04132 ⟶ 2607.09814)	**Negative.** Implemented as specified; it solved 48/48 planted targets on the registered 16–26-bit prime ladder. • **Scaling** in counted F_p ops, against $\log_2 r$: ▸ search phases: slope 0.905 (95% bootstrap 0.769–1.021); ▸ all phases: 0.783 (0.669–0.879); ▸ candidate minors examined: 0.929 (0.793–1.048), with a median of 0.37–2.1 × r per curve. • **Constant:** 260× (215–319) to 6715× (2263–15336) the cost of `rho.negation` per target. • **Falsifier met.** The slope is ≥ 0.5 and the constant is far above rho. • The paper's own Table 2 says it is "no better than exhaustive search".
Hu ePrint 2025/015	**Blocked.** • The full text is behind an IACR Cloudflare challenge, which was not bypassed. • The abstract leaves "IICA" undefined, as well as "integer up/down". • The abstract concerns the DLP in F_p^*, not the ECDLP. • Nothing could be implemented. The original requirement stays unresolved. • The companion ePrint 2024/1923 (abstract) says no subexponential prime-field ECDLP method is known.

The method, in one paragraph

- **Kernel.** Pick $\ell = 3n'$ points $n_i P$ and ℓ points $-n_j Q$. The matrix M of all degree-n' monomials at these 2ℓ points has a left kernel K of dimension ℓ (extra-points theorem).
- **Zero minor ⇒ logarithm.** A maximal zero minor of K is a kernel vector with ℓ zeros. Its support is $3n'$ points on one degree-n' curve, so they sum to O, which gives $\sum n_i - m \sum n_j \equiv 0$.
- **Guess.** 2607.09814 guesses $\ell' = \ell/2$ dense columns, transposes and re-reduces.
- **Determine.** For every set b of $\ell' - d$ dense columns, take the d-dimensional left kernel T. Build the $(\ell' + d) \times d$ signature matrix. Hash the 1-dimensional kernels of all its $(d-1)$-row submatrices; a repeat is a zero $d \times d$ minor.
- **Defect.** d is the least value giving success ≥ 0.9 by the paper's Eq. (4): 3 at 16–24 bits, 4 at 26 bits.

Results

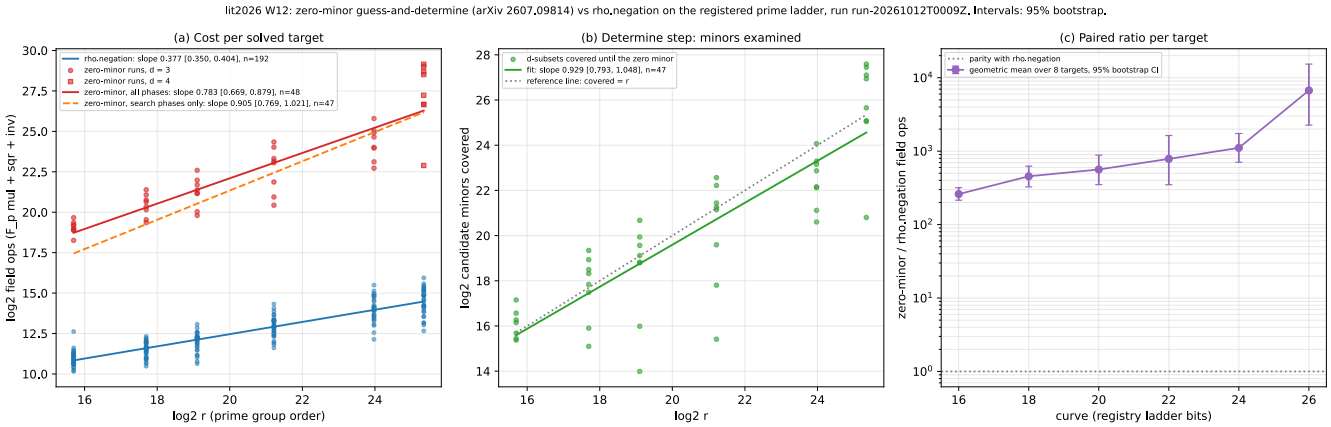


Figure 1: Run-20261012T0009Z, registered ladder `find_prime_order_curve(b, 59297)`, b = 16…26, 8 targets per curve. (a) Counted field ops per solved target, with OLS fits (95% bootstrap). (b) Candidate d-subsets covered, with the line covered = r for reference. (c) Paired per-target ratio of zero-minor to rho.negation (geometric mean, 95% bootstrap).

bits	curve	r	rho.negation median ops	zero-minor median ops	ratio zm/rho (95% bootstrap)
16	`icv1-fp16-t295-8d3c3165`	53117	1.92e3	5.53e5	260 (215–319)
18	`icv1-fp18-tm337-d28d5e09`	212437	2.98e3	1.48e6	455 (326–626)

20	`icv1-fp20-t727-cd198a38`	562427	4.62e3	2.54e6	563 (350–883)
22	`icv1-fp22-tm1385-475dcb5f`	2435809	7.93e3	9.03e6	785 (349–1633)
24	`icv1-fp24-t1059-6df599df`	16482251	1.67e4	2.16e7	1109 (706–1743)
26	`icv1-fp26-tm1775-7e8fb6df`	42653927	2.49e4	2.71e8	6715 (2263–15336)

Identity checks.

- Every one of the 48 zero-minor and 192 rho runs recovered the planted logarithm. Each was checked by $[m]G = Q$ outside the solver.
- A full replay reproduced 240/240 rows with identical counts.
- The defect sweep at 20 bits ($d = 2, 3, 4$, 8 targets each) gave identical counts under two binaries. Medians: 1.21e7, 2.54e6 and 3.32e6 ops.

Interpretation (labelled).

- Each candidate maximal minor is an ℓ-subset of points, which sums to O with probability about $1/r$.
- The signature machinery tests candidates in bulk, but it does not raise that probability. So the work is $\Theta(r)$ candidates.
- The covered/r medians and the search slope near 1 match this.

Ambiguities and what was not implemented

1. **Signature.** The paper's RREF signature equals Tc up to a scalar; the code uses Tc.
2. **Representative of a line.** The code normalises the last non-zero entry to 1; the paper normalises the first. Both are canonical.
3. **Collisions with more than d rows** are skipped, as in Algorithm 2: 0 occurred.
4. **Rank-deficient submatrices** are treated as zero minors: 0 occurred.
5. **After a failed guess a**, the code redraws a. Every run succeeded on its first a.
6. n' is the largest even integer $\leq \log_2 r$.
7. **The "Pollard rho on penultimate intersections" (§5)** defines no walk, so it was not implemented.
8. **Gray-code and rank-one updates** were not implemented. At 26 bits the T computation is only 7.7% of the ops; the $(d-1)$-kernels are 91%.

Limits

- **Sizes.** 16–26 bits ($r \leq 2^{25.35}$), the largest registered ladder curve. The workstream asked for 20–30 bits; 28–30 needs new registered curves (partial).
- **Seeds.** One seed per zero-minor target.
- **Rho's slope.** rho's fitted field-op slope (0.377) is below 0.5 because of the walk's fixed set-up at these sizes.
- **Wall time.** L0 only, not used for any conclusion.
- **Claims.** No deployed-curve or asymptotic claim is made.

Sources: arXiv 1703.07544v3, 2310.04132v1, 2607.09814v2 (full text); ePrint 2025/015 and 2024/1923 (abstract pages only). Code: `examples/lit2026_w12_zero_minor.rs`. Data: `reports/run-20261012T0009Z/`. Details: `README.md`, `HU_EPRINT_2025_015.md`.